SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I A. Sravanthi 119232507 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Sravanthi
Signature of the student:

95/100

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

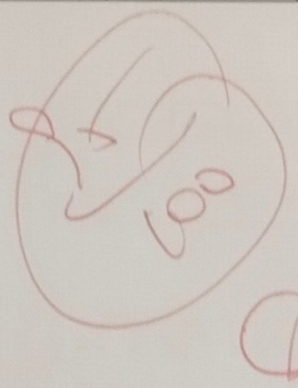
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i.Map how each affects the Data structure and Security level
        ii.Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting 'Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

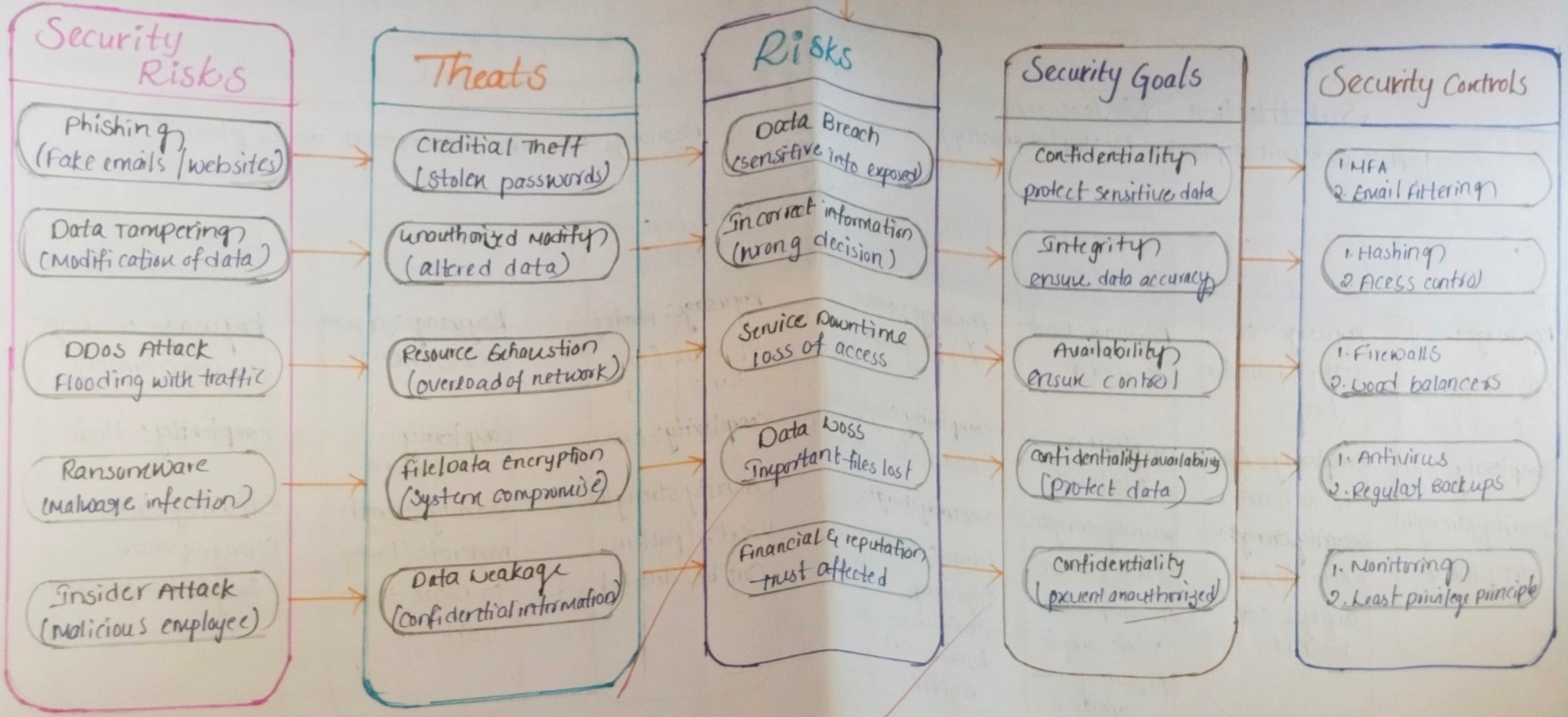
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



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1)

Security Attacks → Threats → Risks → Security Goals



2)

Classical Encryption Schemes

Substitution Techniques & Transposition Techniques

Substitution Techniques

Replace each unit of plaintext (letter/character)

Caesar Cipher

- Key usage: Single Key (shift value 0-25)
- Complexity: Low
- Security strength: Very weak

Monoalphabetic Cipher

- Key usage: -26- letter substitution key
- Complexity: Low to medium
- security strength: weak (frequency analysis can break it)

Vigenere cipher

- Key usage: Repeating keyword
- complexity: Medium
- security strength: stronger than monoalphabetic resists simple frequency analysis

Hill Cipher

- Key usage: Matrix key (keyword)
- complexity: High, Moderate
- Security strength: Strong (resist frequency analysis, depends on matrix)

Transposition Techniques

Rearrange the order of characters in the plaintext

Rail Fence Cipher

- Key usage: Number of rails (ex: 2, 3, 4, 5 - - -)
- Complexity: Low
- Security strength: weak (pattern can be detected)

Columnar Transposition

- Key usage: keyword (ex: ZEBRA)
- Complexity: Medium
- Security strength: Moderate (harder to break than rail)

Double Transposition

- Key usage: Two keys (row + column)
- complexity: High
- Security strength: Stronger than Single transposition

Comparison Table

Feature	Caesar	Monoalphabetic	Vigenere	Hill	Rail Fence	Columnar	Double Transposition
Key usage	Shift value	26-letter substitution	Repeating keyword	Matrix key	Number of rails	Keyword	Two keys
Complexity	Low	Low-medium	Medium	High	Low	Medium	High
Security Strength	Very weak	Weak	Stronger	Strong	Weak	Moderate	Stronger
Data Structure	Same length	Same length	Same length	Same length	Same length	Same length	Same length
Typical Vulnerability	Brute-force	Frequency Analysis	Kasiski test	known plaintext	pattern detection	Frequency analysis	known-plaintext attack

Key Differences

1. Substitution changes the actual characters ; Transposition changes their positions
2. Substitution (Ceaser, Mono, Vignere, Hill) uses keys to replace letters, Transposition)
3. Substitution is more vulnerable to frequency analysis - Transposition

Similarities

1. Both are classical encryption techniques
2. Both keep the data length unchanged
3. Both use a key (even if simple)
4. Both can be combined (ex: Vignere + Transposition) for stronger security).